Week 11 DAM Project

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```
knitr::opts_chunk$set(echo = TRUE)
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.2.0     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.2     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

The task here is to load your Danish Monarchs csv into R using the `tidyverse` toolkit, calculate and explore the kings' duration of reign with pipes `%>%` in `dplyr` and plot it over time.

Load the kings

Make sure to first create an `.Rproj` workspace with a `data/` folder where you place either your own dataset or the provided `kings.csv` dataset.

1. Look at the dataset that are you loading and check what its columns are separated by? (hint: open it in plain text editor to see)

List what is the

separator: "reign_kings.csv" is seperated by "_"

2. Create a `kings` object in R with the different functions below and inspect the different outputs.

- `read.csv()`
- `read_csv()`
- `read.csv2()`
- `read_csv2()`

```
# FILL IN THE CODE BELOW and review the outputs
kings1 <- read.csv("data/kings.csv")

kings2 <- read_csv("data/kings.csv")
```

```
## Rows: 57 Columns: 11
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr (11): Name, House, Start_year, End_year, Birth_year, Death_year, Gender,...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
kings3 <- read.csv2("data/kings.csv")

kings4 <- read_csv2("data/kings.csv")
```

```
## ℹ Using "','" as decimal and "'.'" as grouping mark. Use `read_delim()` for more control.
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 57 Columns: 1
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ";"
## chr (1): Name,House,Start_year,End_year,Birth_year,Death_year,Gender,Dynasty...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Answer: 1. Which of these functions is a `tidyverse` function? Read data with it below into a `kings` object # read_csv 2. What is the result of running `class()` on the `kings` object created with a tidyverse function. #“spec_tbl_df” “tbl_df” “tbl” “data.frame” 3. How many columns does the object have when created with these different functions? # 1 or 6 columns 4. Show the dataset so that we can see how R interprets each column

kings

```
# COMPLETE THE BLANKS BELOW WITH YOUR CODE, then turn the 'eval' flag in this chunk to TRUE.

kings <- read_csv("data/kings.csv", na = c("NULL",""))
```

```
## Rows: 57 Columns: 11
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr (7): Name, House, Gender, Dynasty, Source, BirthDMY, DeathDMY
## dbl (4): Start_year, End_year, Birth_year, Death_year
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
class(kings)
```

```
## [1] "spec_tbl_df" "tbl_df"      "tbl"         "data.frame"
```

```
glimpse(kings)
```

```
## Rows: 57
## Columns: 11
## $ Name       <chr> "Gorm den Gamle", "Harald 1. Blåtand", "Toke_Gormsen", "Sve…
## $ House      <chr> "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Fa…
## $ Start_year <dbl> 936, NA, 985, NA, 1014, 1018, 1035, 1042, 1047, 1074, 1080,…
## $ End_year   <dbl> 958, NA, 986, NA, 1018, 1035, 1042, 1047, 1074, 1080, 1086,…
## $ Birth_year <dbl> 908, 936, NA, NA, 963, NA, 995, 1018, 1024, 1019, 1041, 104…
## $ Death_year <dbl> 958, 985, 986, 1014, NA, 1035, 1042, 1047, 1076, 1080, 1086…
## $ Gender     <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M",…
## $ Dynasty    <chr> "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet",…
## $ Source     <chr> "https://da.wikipedia.org/wiki/Gorm_den_Gamle", "https://da…
## $ BirthDMY   <chr> "908", "936", NA, "17/04/963", "994", "995", "1018", "1024"…
## $ DeathDMY   <chr> "958", "987", NA, "03/02/1014", "1018", "12/11/1035", "08/0…
```

```
head(kings)
```

```
## # A tibble: 6 × 11
##   Name     House Start_year End_year Birth_year Death_year Gender Dynasty Source
##   <chr>    <chr>      <dbl>    <dbl>      <dbl>      <dbl> <chr>  <chr>   <chr>
## 1 Gorm de… Gorm         936      958        908        958 M      Jellin… https…
## 2 Harald … Gorm          NA       NA        936        985 M      Jellin… https…
## 3 Toke_Go… Gorm         985      986         NA        986 M      Jellin… https…
## 4 Svend 1… Gorm          NA       NA         NA       1014 M      Jellin… https…
## 5 Harald … Gorm        1014     1018        963         NA M      Jellin… https…
## 6 Knud 1.… Gorm        1018     1035         NA       1035 M      Jellin… https…
## # ℹ 2 more variables: BirthDMY <chr>, DeathDMY <chr>
```

Calculate the duration of reign for all the kings in your table

You can calculate the duration of reign in years with `mutate` function by subtracting the equivalents of your `startReign` from `endReign` columns and writing the result to a new column called `duration`. But first you need to check a few things:

- Is your data messy? Fix it before re-importing to R
- Do your start and end of reign columns contain NAs? Choose the right strategy to deal with them: `na.omit()`, `na.rm=TRUE`, `!is.na()`

Create a new column called `duration` in the kings dataset, utilizing the `mutate()` function from tidyverse. Check with your group to brainstorm the options.

```
# YOUR CODE

kings %>%
  select(Start_year, End_year) %>%
  mutate(duration = End_year - Start_year)
```

```
## # A tibble: 57 × 3
##    Start_year End_year duration
##         <dbl>    <dbl>    <dbl>
##  1        936      958       22
##  2         NA       NA       NA
##  3        985      986        1
##  4         NA       NA       NA
##  5       1014     1018        4
##  6       1018     1035       17
##  7       1035     1042        7
##  8       1042     1047        5
##  9       1047     1074       27
## 10       1074     1080        6
## # ℹ 47 more rows
```

Calculate the average duration of reign for all rulers

Do you remember how to calculate an average on a vector object? If not, review the last two lessons and remember that a column is basically a vector. So you need to subset your `kings` dataset to the `duration` column. If you subset it as a vector you can calculate average on it with `mean()` base-R function. If you subset it as a tibble, you can calculate average on it with `summarize()` tidyverse function. Try both ways!

- You first need to know how to select the relevant `duration` column. What are your options?
- Is your selected `duration` column a tibble or a vector? The `mean()` function can only be run on a vector. The `summarize()` function works on a tibble.
- Are you getting an error that there are characters in your column? Coerce your data to numbers with `as.numeric()`.
- Remember to handle NAs: `mean(X, na.rm=TRUE)`

```
kings %>%
   select(Start_year, End_year) %>%
  mutate(duration = End_year - Start_year) %>%
  summarise(duration = mean(duration, na.rm = TRUE))
```

```
## # A tibble: 1 × 1
##   duration
##      <dbl>
## 1     19.6
```

How many and which kings enjoyed a longer-than-average duration of reign?

You have calculated the average duration above. Use it now to `filter()` the `duration` column in `kings` dataset. Display the result and also count the resulting rows with `count()`

```
kings %>%
  select(Name, Start_year, End_year) %>%
  mutate(duration = End_year - Start_year) %>%
  filter(duration > mean(duration, na.rm = TRUE))
```

```
## # A tibble: 25 × 4
##    Name                  Start_year End_year duration
##    <chr>                      <dbl>    <dbl>    <dbl>
##  1 Gorm den Gamle               936      958       22
##  2 Svend 2. Estridsen          1047     1074       27
##  3 Niels                       1104     1134       30
##  4 Valdemar 1. den Store       1157     1182       25
##  5 Knud 4.                     1182     1202       20
##  6 Valdemar 2. Sejr            1202     1241       39
##  7 Erik 5. Klipping            1259     1286       27
##  8 Erik 6. Menved              1286     1319       33
##  9 Valdemar 4. Atterdag        1340     1375       35
## 10 Erik 7. af Pommern          1396     1439       43
## # ℹ 15 more rows
```

How many days did the three longest-ruling monarchs rule?

- Sort kings by reign `duration` in the descending order. Select the three longest-ruling monarchs with the `slice()` function
- Use `mutate()` to create `Days` column where you calculate the total number of days they ruled
- BONUS: consider the transition year (with 366 days) in your calculation!

```
kings %>%
  select(Name, Start_year, End_year) %>%
  mutate(duration = End_year - Start_year) %>%
  arrange(desc(duration)) %>%
  slice(1:3) %>%
  mutate(Days = duration * 365)
```

```
## # A tibble: 3 × 5
##   Name               Start_year End_year duration  Days
##   <chr>                   <dbl>    <dbl>    <dbl> <dbl>
## 1 Christian 4.             1588     1648       60 21900
## 2 Margrete 2.              1972     2023       51 18615
## 3 Erik 7. af Pommern       1396     1439       43 15695
```

Challenge: Plot the kings' duration of reign through time

What is the long-term trend in the duration of reign among Danish monarchs? How does it relate to the historical violence trends ?

- Try to plot the duration of reign column in `ggplot` with `geom_point()` and `geom_smooth()`
- In order to peg the duration (which is between 1-99) somewhere to the x axis with individual centuries, I recommend creating a new column `midyear` by adding to `startYear` the product of `endYear` minus the `startYear` divided by two (`startYear + (endYear-startYear)/2`).
- Now you can plot the kings dataset, plotting `midyear` along the x axis and `duration` along y axis
- BONUS: add a title, nice axis labels to the plot and make the theme B&W and font bigger to make it nice and legible!

```
# YOUR CODE
```

And to submit this rmarkdown, knit it into html. But first, clean up the code chunks, adjust the date, rename the author and change the `eval=FALSE` flag to `eval=TRUE` so your script actually generates an output. Well done!

library(tidyverse) library(tidyverse)